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ОТЧЁТ
О ЛАБОРАТОРНОЙ РАБОТЕ № 5
Администрирование систем и сетей
по теме:
Конфигурирование основных сетевых служб
Часть 2:
Конфигурирование DHCP
Желаемая оценка: 3

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СОДЕРЖАНИЕ

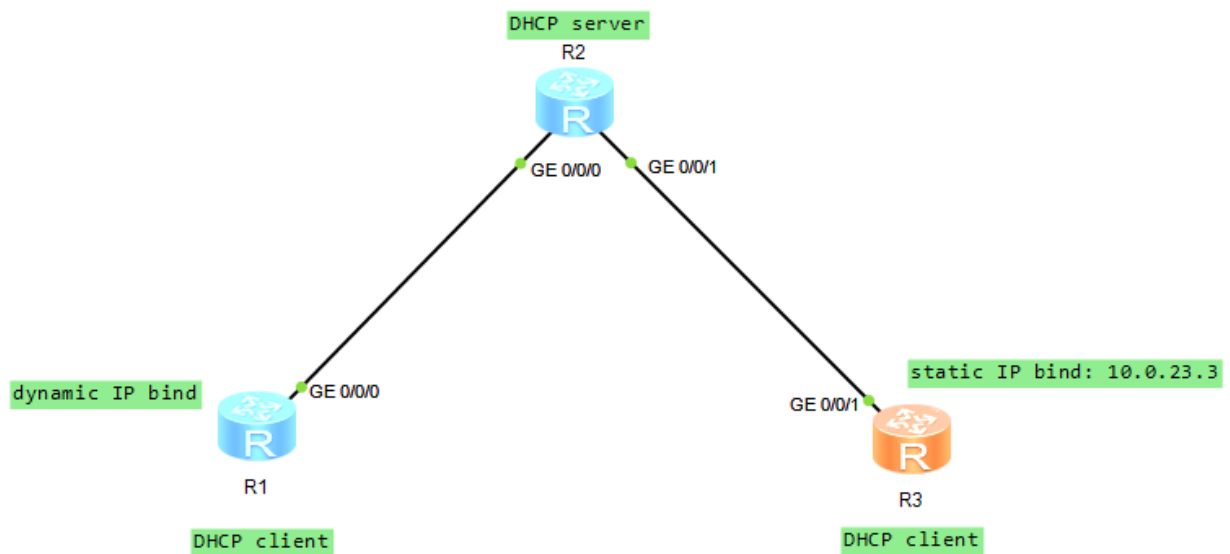
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ПЛАН РАБОТЫ

1. Настройка DHCP-сервера.
2. Настройка DHCP-клиентов.

1 Ход работы

1.1 Настройка основных параметров устройств



```
[Huawei]sysname R1
[R1]
```

```
[Huawei]sysname R2
[R2]
```

```
[Huawei]sysname R3
[R3]
```

Настройка адресов интерфейсов

```
[R2]interface GigabitEthernet 0/0/0
[R2-GigabitEthernet0/0/0]ip address 10.0.12.2 24
[R2-GigabitEthernet0/0/0]quit
[R2]interface GigabitEthernet 0/0/1
[R2-GigabitEthernet0/0/1]ip address 10.0.23.2 24
[R2-GigabitEthernet0/0/1]
```

1.2 Включение функции DHCP

```
[R1]dhcp enable
Info: The operation may take a few seconds. Please wait for a moment.done.
[R1]
```

```
[R2]dhcp enable
Info: The operation may take a few seconds. Please wait for a moment.done.
[R2]
```

```
[R3]dhcp enable
Info: The operation may take a few seconds. Please wait for a moment.done.
[R3]
```

1.3 Настройка пула адресов

Интерфейсная настройка DHCP

```
[R2]interface GigabitEthernet 0/0/0
[R2-GigabitEthernet0/0/0]dhcp select interface
[R2-GigabitEthernet0/0/0]dhcp server dns-list 10.0.12.2
[R2-GigabitEthernet0/0/0]quit
```

Настройка глобального пула

```
[R2]ip pool GlobalPool
Info: It's successful to create an IP address pool.
[R2-ip-pool-GlobalPool]network 10.0.23.0 mask 24
[R2-ip-pool-GlobalPool]dns-list 10.0.23.2
[R2-ip-pool-GlobalPool]gateway-list 10.0.23.2
[R2-ip-pool-GlobalPool]lease day 2 hour 2
[R2-ip-pool-GlobalPool]
```

Узнаем MAC GE 0/0/1 на R3

```
[R3]display interface GigabitEthernet0/0/1
GigabitEthernet0/0/1 current state : UP
Line protocol current state : DOWN
Description:HUAWEI, AR Series, GigabitEthernet0/0/1 Interface
Route Port,The Maximum Transmit Unit is 1500
Internet protocol processing : disabled
IP Sending Frames' Format is PKTFMT_ETHNT_2, Hardware address is 00e0-fce3-303c
Last physical up time    : 2025-11-06 16:13:33 UTC-08:00
Last physical down time : 2025-11-06 16:13:21 UTC-08:00
Current system time: 2025-11-06 16:25:52-08:00
Port Mode: COMMON COPPER
Speed : 1000,  Loopback: NONE
Duplex: FULL,  Negotiation: ENABLE
Mdi    : AUTO
Last 300 seconds input rate 0 bits/sec, 0 packets/sec
Last 300 seconds output rate 0 bits/sec, 0 packets/sec
Input peak rate 96 bits/sec,Record time: 2025-11-06 16:16:27
Output peak rate 0 bits/sec,Record time: -
```

```

Input:  1 packets, 60 bytes
  Unicast:      0,  Multicast:      0
  Broadcast:    1,  Jumbo:          0
  Discard:      0,  Total Error:    0

  CRC:          0,  Giants:         0
  Jabbers:      0,  Throttles:      0
  Runts:        0,  Symbols:        0
  Ignoreds:     0,  Frames:         0

Output:  0 packets, 0 bytes
  Unicast:      0,  Multicast:      0
  Broadcast:    0,  Jumbo:          0
  Discard:      0,  Total Error:    0

  Collisions:   0,  ExcessiveCollisions: 0
  Late Collisions: 0,  Deferreds:      0

  Input bandwidth utilization threshold : 100.00%
  Output bandwidth utilization threshold: 100.00%
  Input bandwidth utilization   :    0%
  Output bandwidth utilization  :    0%

```

[R3]

1.4 Включение функции DHCP-сервера

Настройка статической привязки IP к MAC, R3 на интерфейсе GE 0/0/1
получает постоянный IP

```
[R2-ip-pool-GlobalPool]static-bind ip-address 10.0.23.3 mac-address 00e0-
fce3-303c
```

Настройка выдачи IP из глобального пула для интерфейса GE0/0/1 на
R2

```
[R2]interface GigabitEthernet 0/0/1
[R2-GigabitEthernet0/0/1]dhcp select global
[R2-GigabitEthernet0/0/1]
```

1.5 Настройка DHCP-клиентов

```
[R1]interface GigabitEthernet 0/0/0
[R1-GigabitEthernet0/0/0]ip address dhcp-alloc
```

```
[R3]interface GigabitEthernet 0/0/1
[R3-GigabitEthernet0/0/1]ip address dhcp-alloc
```

Вывод IP адресов маршрутизаторов

```
[R1]display ip interface brief
*down: administratively down
^down: standby
(l): loopback
(s): spoofing
The number of interface that is UP in Physical is 2
The number of interface that is DOWN in Physical is 2
The number of interface that is UP in Protocol is 2
The number of interface that is DOWN in Protocol is 2
```

Interface	IP Address/Mask	Physical	Protocol
GigabitEthernet0/0/0	10.0.12.254/24	up	up
GigabitEthernet0/0/1	unassigned	down	down
GigabitEthernet0/0/2	unassigned	down	down
NULL0	unassigned	up	up(s)

```
[R1]
```

```
[R1]display dns server
```

```
Type:
D:Dynamic      S:Static
```

No.	Type	IP Address
1	D	2.12.0.10

Шлюз по умолчанию

```
[R1]display ip routing-table
Route Flags: R - relay, D - download to fib
```

```
-----
Routing Tables: Public
```

```
Destinations : 8          Routes : 8
```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
0.0.0.0/0	Unr	60	0	D	10.0.12.2	GigabitEthernet0/0/0
10.0.12.0/24	Direct	0	0	D	10.0.12.254	GigabitEthernet0/0/0
10.0.12.254/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

255.255.255.255/32 Direct 0 0 D 127.0.0.1 InLoopBack0

[R1]

[R3]display ip interface brief

*down: administratively down

^down: standby

(l): loopback

(s): spoofing

The number of interface that is UP in Physical is 2

The number of interface that is DOWN in Physical is 2

The number of interface that is UP in Protocol is 2

The number of interface that is DOWN in Protocol is 2

Interface	IP Address/Mask	Physical	Protocol
GigabitEthernet0/0/0	unassigned	down	down
GigabitEthernet0/0/1	10.0.23.3/24	up	up
GigabitEthernet0/0/2	unassigned	down	down
NULL0	unassigned	up	up(s)

[R3]

[R3]display dns server

Type:

D:Dynamic S:Static

No.	Type	IP Address
1	D	2.23.0.10

Шлюз по умолчанию

[R3]display ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 8

Routes : 8

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
0.0.0.0/0	Unr	60	0	D	10.0.23.2	GigabitEthernet
0/0/1						
10.0.23.0/24	Direct	0	0	D	10.0.23.3	GigabitEthernet
0/0/1						
10.0.23.3/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/1						
10.0.23.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/1						
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

[R3]

Информация о назначении IP адресов в глобальном пуле

[R2]display ip pool name GlobalPool

Pool-name : GlobalPool
Pool-No : 1
Lease : 2 Days 2 Hours 0 Minutes
Domain-name : -
DNS-server0 : 10.0.23.2
NBNS-server0 : -
Netbios-type : -
Position : Local Status : Unlocked
Gateway-0 : 10.0.23.2
Mask : 255.255.255.0
VPN instance : --

	Start	End	Total	Used	Idle(Expired)	Conflict
Disable						
	10.0.23.1	10.0.23.254	253	1	252(0)	0

[R2]

Информация о настройках пула конкретного интерфейса

[R2]display ip pool interface GigabitEthernet0/0/0

Pool-name : GigabitEthernet0/0/0
Pool-No : 0
Lease : 1 Days 0 Hours 0 Minutes
Domain-name : -
DNS-server0 : 10.0.12.2
NBNS-server0 : -
Netbios-type : -
Position : Interface Status : Unlocked
Gateway-0 : 10.0.12.2
Mask : 255.255.255.0
VPN instance : --

	Start	End	Total	Used	Idle(Expired)	Conflict
Disable						
	10.0.12.1	10.0.12.254	253	1	252(0)	0

[R2]

Конфигурации устройств

```
<R1>display current-configuration
[V200R003C00]
#
 sysname R1
#
 snmp-agent local-engineid 800007DB03000000000000
 snmp-agent
#
 clock timezone China-Standard-Time minus 08:00:00
#
 portal local-server load portalpage.zip
#
 drop illegal-mac alarm
#
 set cpu-usage threshold 80 restore 75
#
 dhcp enable
#
 aaa
 authentication-scheme default
 authorization-scheme default
 accounting-scheme default
 domain default
 domain default_admin
 local-user admin password cipher %$$K8m.Nt84DZ}e#<0`8bmE3Uw}%$$
 local-user admin service-type http
#
 firewall zone Local
 priority 15
#
 interface GigabitEthernet0/0/0
 ip address dhcp-alloc
#
 interface GigabitEthernet0/0/1
#
 interface GigabitEthernet0/0/2
#
 interface NULL0
#
 user-interface con 0
 authentication-mode password
 user-interface vty 0 4
 user-interface vty 16 20
#
 wlan ac
#
return
```

```
[R2]display current-configuration
[V200R003C00]
#
 sysname R2
#
 snmp-agent local-engineid 800007DB03000000000000
 snmp-agent
#
 clock timezone China-Standard-Time minus 08:00:00
#
portal local-server load portalpage.zip
#
 drop illegal-mac alarm
#
 set cpu-usage threshold 80 restore 75
#
dhcp enable
#
ip pool GlobalPool
 gateway-list 10.0.23.2
 network 10.0.23.0 mask 255.255.255.0
 static-bind ip-address 10.0.23.3 mac-address 00e0-fce3-303c
 lease day 2 hour 2 minute 0
 dns-list 10.0.23.2
#
aaa
 authentication-scheme default
 authorization-scheme default
 accounting-scheme default
 domain default
 domain default_admin
 local-user admin password cipher %$$K8m.Nt84DZ}e#<0`8bmE3Uw}%$$
 local-user admin service-type http
#
firewall zone Local
 priority 15
#
interface GigabitEthernet0/0/0
 ip address 10.0.12.2 255.255.255.0
 dhcp select interface
 dhcp server dns-list 10.0.12.2
#
interface GigabitEthernet0/0/1
 ip address 10.0.23.2 255.255.255.0
 dhcp select global
#
interface GigabitEthernet0/0/2
#
interface NULL0
#
user-interface con 0
 authentication-mode password
```

```
user-interface vty 0 4
user-interface vty 16 20
#
wlan ac
#
return
```

```
<R3>display current-configuration
[V200R003C00]
#
 sysname R3
#
 snmp-agent local-engineid 800007DB0300000000000000
 snmp-agent
#
 clock timezone China-Standard-Time minus 08:00:00
#
 portal local-server load portalpage.zip
#
 drop illegal-mac alarm
#
 set cpu-usage threshold 80 restore 75
#
 dhcp enable
#
 aaa
 authentication-scheme default
 authorization-scheme default
 accounting-scheme default
 domain default
 domain default_admin
 local-user admin password cipher %$$K8m.Nt84DZ}e#<0`8bmE3Uw}%$$
 local-user admin service-type http
#
 firewall zone Local
 priority 15
#
 interface GigabitEthernet0/0/0
#
 interface GigabitEthernet0/0/1
 ip address dhcp-alloc
#
 interface GigabitEthernet0/0/2
#
 interface NULL0
#
 user-interface con 0
 authentication-mode password
 user-interface vty 0 4
 user-interface vty 16 20
#
 wlan ac
#
```

return

ЗАКЛЮЧЕНИЕ

Цель работы достигнута.